

Homework 4

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Exercise 1

Let B be the event that Tweety is a bird, F be the event that Tweety flies, P be the event that Tweety is a penguin, and I be the event that Tweety is the superhero iron penguin. We then have that:

$$\begin{aligned}P(F|B) &= \frac{P(F, B)}{P(B)} = 0.8 \\P(F|B, P) &= \frac{P(F, B, P)}{P(B, P)} = 0.2 \\P(F|B, P, I) &= \frac{P(F, B, P, I)}{P(B, P, I)} = 0.8\end{aligned}$$

To simplify things, let $I \subseteq P \subseteq B$. Then:

$$\begin{aligned}P(F|B) &= \frac{P(F, B)}{P(B)} = 0.8 \\P(F|B, P) &= \frac{P(F, P)}{P(P)} = 0.2 \\P(F|B, P, I) &= \frac{P(F, I)}{P(I)} = 0.8\end{aligned}$$

We then arbitrarily let $P(B) = 1$. Then:

$$\begin{aligned}P(F|B) &= \frac{P(F)}{1} = 0.8 \\P(F) &= 0.8\end{aligned}$$

If $P(P) > 0.2$, there will be a nonzero lower bound on $P(F, P)$. To avoid this, we let $P(P) = 0.2$. Then:

$$P(F|B, P) = \frac{P(F, P)}{0.2} = 0.2$$

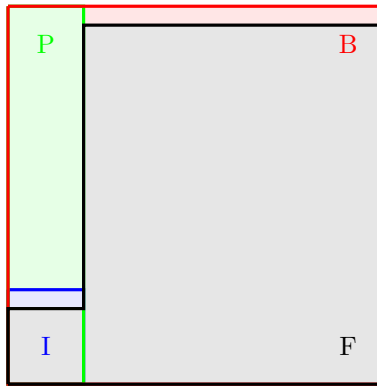
$$P(F, P) = 0.04$$

To simplify things further, assume Iron Penguin is the only penguin that can fly. Then, $P(F, I) = P(F, P)$, and so:

$$P(F|B, P, I) = \frac{0.04}{P(I)} = 0.8$$

$$P(I) = 0.05$$

Visualizing this using a Venn diagram:



Exercise 2

From the definition of conditional probability, for $P(B) \neq 0$:

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

Then:

$$P(A|B) \cdot P(B) = \frac{P(A, B)}{P(B)} \cdot P(B)$$

$$P(A|B) \cdot P(B) = P(A, B)$$

Exercise 3

Repeatedly applying the rule derived in Exercise 2:

$$\begin{aligned}P(A, B, C, D) &= P(A|B, C, D) \cdot P(B, C, D) \\&= P(A|B, C, D) \cdot P(B|C, D) \cdot P(C, D) \\&= P(A|B, C, D) \cdot P(B|C, D) \cdot P(C|D) \cdot P(D)\end{aligned}$$

Exercise 4

Let E_0 be the event that variable X_i takes value 0 and E_1 be the event that variable X_i takes value 1.

$$E_0 = \{x_i \in \Omega \mid x_i = 0\}$$

$$E_1 = \{x_i \in \Omega \mid x_i = 1\}$$

Since E_0 and E_1 comprise the whole sample space ($E_0 \cup E_1 = \Omega$):

$$P(E) = P((E \cap E_0) \cup (E \cap E_1))$$

Then, since E_0 and E_1 are mutually disjoint ($E_0 \cap E_1 = \emptyset$):

$$P(E) = P(E \cap E_0) + P(E \cap E_1)$$

Using the definitions of E_0 and E_1 :

$$P(E) = \sum_{x \in E_0} P(E, X_i = x) + \sum_{x \in E_1} P(E, X_i = x)$$

Since x can only be in either E_0 or E_1 :

$$P(E) = \sum_x P(E, X_i = x)$$

Exercise 5

Repeatedly applying the rule derived in Exercise 4:

$$\begin{aligned} P(X = 1, Y = 0) &= \sum_u P(X = 1, Y = 0, U = u) \\ &= \sum_{u,v} P(X = 1, Y = 0, U = u, V = v) \\ &= \sum_{u,v,w} P(X = 1, Y = 0, U = u, V = v, W = w) \end{aligned}$$

Exercise 6

Suppose we have events A , B , and C such that $P(B, C) \neq 0$ and A and B are independent given C . Then:

$$\begin{aligned} P(A|B, C) &= \frac{P(A, B, C)}{P(B, C)} \\ &= \frac{P(A, B|C) \cdot P(C)}{P(B, C)} \\ &= \frac{P(A|C) \cdot P(B|C) \cdot P(C)}{P(B, C)} \\ &= \frac{P(A|C) \cdot P(B, C)}{P(B, C)} \\ &= P(A|C) \end{aligned}$$